

10.4. Prove the following theorem using the theorem of deduction and its reverse.

- proof only using axioms! -

$$\vdash (p \wedge q \rightarrow r) \rightarrow (p \rightarrow (q \rightarrow r))$$

Theorem of deduction

If $U_1, \dots, U_{m-1}, U_m \vdash V$ then
 $U_1, \dots, U_{m-1} \vdash U_m \rightarrow V$

Reverse of theorem of deduction

If $U_1, \dots, U_{m-1} \vdash U_m \rightarrow V$ then
 $U_1, \dots, U_{m-1} \vdash V$

Step 1: We apply the reverse of the theorem of deduction.

If $\vdash (p \wedge q \rightarrow r) \rightarrow (p \rightarrow (q \rightarrow r))$ then

If $p \wedge q \rightarrow r \vdash p \rightarrow (q \rightarrow r)$ then

If $p \wedge q \rightarrow r, p \vdash q \rightarrow r$ then

$p \wedge q \rightarrow r, p, q \vdash r$ (*)

We could skip the first step and guess this deduction

Step 2: We prove (*) deduction.

f1: $p \wedge q \rightarrow r$

f2: p

f3: q

f4: $p \wedge q = f2 \wedge f3 \rightarrow$ conjunction in conclusion

$f2, f4 \vdash_{mp} r: f5$

Step 3: We apply the theorem of deduction:

if $p \wedge q \rightarrow r, p, q \vdash r$ then

if $p \wedge q \rightarrow r, p \vdash q \rightarrow r$ then

if $\vdash (p \wedge q \rightarrow r) \rightarrow (p \rightarrow (q \rightarrow r))$ then

if $\vdash p \wedge q \rightarrow r \vdash p \rightarrow (q \rightarrow r)$, then

$T_1: \vdash (p \wedge q \rightarrow r) \rightarrow (p \rightarrow (q \rightarrow r))$ theorem

we can permute them $\rightarrow 3!$ theorems.

$T_2: \vdash (p \wedge q \rightarrow r) \rightarrow (q \rightarrow (p \rightarrow r))$

$T_3: \vdash p \rightarrow ((p \wedge q \rightarrow r) \rightarrow (q \rightarrow r))$

$T_4: \vdash p \rightarrow (q \rightarrow (p \wedge q \rightarrow r) \rightarrow r)$

$T_5: \vdash q \rightarrow (p \rightarrow (p \wedge q \rightarrow r) \rightarrow r)$

$T_6: \vdash q \rightarrow ((p \wedge q \rightarrow r) \rightarrow (p \rightarrow r))$

Predicate logic

1.4. Transform from natural language into predicate formulas.

If x and y are positive prime numbers, $x \neq y$, $x \neq 2$, $y \neq 2$,
their sum and difference are even numbers and their product is an odd nr.

$D = \mathbb{N}$

$2 \in D$

2 constant.

Predicate symbols: prime, even, pos.

prime: $D \rightarrow \{T, F\}$, prime(x) = T if x is prime.

even: $D \rightarrow \{T, F\}$, even(x) = T if x is even

(odd $\rightarrow$ $\neg$ even)

eg: $D \times D \rightarrow \{T, F\}$, eg(x, y) = T if $x = y$

pos: $D \rightarrow \{T, F\}$, pos(x) = T if $x > 0$.

Function symbols: sum, diff, product.

sum: $D \times D \rightarrow D$, sum(x, y) = $x + y$

diff: $D \times D \rightarrow D$, diff(x, y) = $x - y$

prod: $D \times D \rightarrow D$, prod(x, y) = $x \cdot y$

Translation:

$(\forall x), (\forall y) (\text{pos}(x) \wedge \text{prime}(x) \wedge \text{pos}(y) \wedge \text{prime}(y) \wedge \neg \text{eq}(x, y) \wedge \neg \text{eq}(x, z) \wedge \neg \text{eq}(y, z) \rightarrow$
 $\rightarrow \text{even}(\text{sum}(x, y)) \wedge \text{even}(\text{diff}(x, y)) \wedge \neg \text{even}(\text{prod}(x, y)).$

Properties of eq:

$(\forall) x, \text{eq}(x, x) \rightarrow \text{reflexivity}$

$(\forall) x, (\forall) y, (\text{eq}(x, y) \rightarrow \text{eq}(y, x)) \rightarrow \text{symmetry}$

$(\forall) x, (\forall) y, (\forall) z, \text{eq}(x, y) \wedge \text{eq}(y, z) \rightarrow \text{eq}(x, z) \rightarrow \text{transitivity}$

Properties of sum:

$x + y = y + x \text{ (comm.)}$

$x + 0 = x \text{ (e.m.)}$

$x + (y + z) = (x + y) + z \text{ (trans.)}$

$x * (y + z) = x * y + x * z \text{ (distrib.)}$

$(\forall) x, (\forall) y, \text{eq}(\text{sum}(x, y), \text{sum}(y, x)) \rightarrow \text{comm}$

$(\forall) x, \text{eq}(\text{sum}(x, 0), x) \rightarrow \text{e.m. 0}$

$(\forall) x, (\forall) y, (\forall) z, \text{eq}(\text{sum}(x, \text{sum}(y, z)), \text{sum}(\text{sum}(x, y), z)) \rightarrow \text{trans}$

$(\forall) x, (\forall) y, (\forall) z, \text{eq}(\text{prod}(x, \text{sum}(y, z)), \text{sum}(\text{prod}(x, y), \text{prod}(x, z))) \rightarrow \text{distrib.}$

3.4. Use deductions and inference rules:

H_1 : If x is the king and y is his eldest son, then y can become a king.

H_2 : If x is the king and y defeats x, then y will become the king.

H_3 : ^{R_3}
Richard III is the king.

H_4 : ^{$He7$}
Henry VII defeated Richard III

H_5 : ^{$He8$}
Henry VIII is Henry VII's eldest son.

C: Can Henry VIII become the king?

Inference rules:

$u, v \vdash u \wedge v$
conj.

$u, u \rightarrow v \vdash v$

$u, x \vdash \text{mp}$
 $(\forall x), U(x) \vdash \text{univ. inst.} \quad U(t), t\text{-term.}$

$$H_1, \dots, H_5 \vdash ? C$$

Constants: $R_3, \text{He7}, \text{He8}$

Predicate symbols:

$\text{King}(x) = \text{T}$ if x is King

$\text{Os}(x, y) = \text{T}$ if y is x 's oldest son.

$\text{D}(x, y) = \text{T}$ if y defeats x .

Translation:

$$H_1: (\forall x), (\forall y), \text{King}(x) \wedge \text{Os}(x, y) \rightarrow \text{King}(y)$$

$$H_2: (\forall x), (\forall y), \text{King}(x) \wedge \text{D}(x, y) \rightarrow \text{King}(y)$$

$$H_3: \text{King}(R_3) \quad (\text{it's a fact.})$$

$$H_4: \text{D}(R_3, \text{He7})$$

$$H_5: \text{Os}(\text{He7}, \text{He8})$$

$$C: \text{King}(\text{He8})$$

$$H_2 \vdash \text{univ. inst.} \quad (\forall y), \text{King}(R_3) \wedge \text{D}(R_3, y) \rightarrow \text{King}(y) \quad (\text{f6})$$

$x \leftarrow R_3$

(we replace x with R_3 through univ. inst.)

$$\text{f6} \vdash \text{univ. inst.} \quad \text{King}(R_3) \wedge \text{D}(R_3, \text{He7}) \rightarrow \text{King}(\text{He7}) \quad (\text{f7})$$

$y \leftarrow \text{He7}$

$$H_3, H_4 \vdash \text{conj.} \quad \text{King}(R_3) \wedge \text{D}(R_3, \text{He7}) \quad (\text{f8})$$

$$\text{f7}, \text{f8} \vdash \text{mp} \quad \text{King}(\text{He7}) \quad (\text{f9})$$

$$H_1 \vdash \text{univ. inst.} \quad (\forall y), \text{King}(\text{He7}) \wedge \text{Os}(\text{He7}, y) \rightarrow \text{King}(y) \quad (\text{f10})$$

$$\text{f9} \vdash \text{univ. inst.} \quad \text{King}(\text{He7}) \wedge \text{Os}(\text{He7}, \text{He8}) \rightarrow \text{King}(\text{He8}) \quad (\text{f11})$$

$y \leftarrow \text{He8}$

$$\text{f9}, \text{f11} \vdash \text{mp} \quad \text{King}(\text{He7}) \wedge \text{Os}(\text{He7}, \text{He8}) \quad (\text{f12})$$

$\neg \neg, \neg \neg \vdash \text{conj.}$

$$\neg 12, \neg 10 \vdash_{\text{mp}} \text{King}(\text{He8}) (C)$$

$\Rightarrow$ The sequence of formulas from $(H_1, \dots, H_5, \neg 6, \dots, \neg 12, C)$ is the deduction (proof) of C from the hypothesis $\{H_1, \dots, H_5\}$.

2.4. Natural language $\Rightarrow$ predicate formula.

All Mary's relatives live in Cluj-Napoca, only her cousin John lives in Bucharest.

D_1 = set of all persons

Mary, John $\in D_1 \rightarrow$ constants.

D_2 = set of all cities

Cluj-Napoca, Bucharest $\in D_2$, constants.

Predicate symbols

$\text{lives} : D_1 \times D_2 \rightarrow \{T, F\}$, $\text{lives}(x, y) = T$ if x lives in y .

$\text{rel} : D_1 \times D_1 \rightarrow \{T, F\}$, $\text{rel}(x, y) = T$ if x related to y .

$\text{cousin} : D_1 \times D_1 \rightarrow \{T, F\}$, $\text{cousin}(x, y) = T$ if x cousin of y .

Symm.
property

Symmetric relation! $\rightarrow$ property of binary relation.

$\wedge (\text{cousin}(x, \text{Mary}) \wedge \neg \text{egu}(x, \text{John})) \rightarrow \text{lives}(x, \text{Cluj}) \wedge (\text{cousin}(\text{John}, \text{Mary}))$

? $\rightarrow \text{lives}(\text{John}, \text{Buc})$

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$(\forall) \text{cousin}(x) \rightarrow \text{relative}(x)$.

